

5.8 Classwork: Calculus and Statistics (no calculator) — Markscheme

1. Let $f(x) = x^3 - 3x + 2$.

- (a) Find $f'(x)$. [1]

$$f'(x) = 3x^2 - 3 \quad (\mathbf{A1})$$

- (b) Find the gradient of the graph of f at $x = 2$. [2]

$$f'(2) = 3(4) - 3 \quad (\mathbf{M1})$$

$$= 9 \quad (\mathbf{A1})$$

- (c) Find the equation of the tangent line to the graph of f at $x = 2$. Give your answer in the form $y = mx + c$. [3]

$$f(2) = 8 - 6 + 2 = 4 \quad (\mathbf{A1})$$

$$y - 4 = 9(x - 2) \quad (\mathbf{M1})$$

$$y = 9x - 14 \quad (\mathbf{A1})$$

- (d) Find the coordinates of the stationary points of f . [2]

$$3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \quad (\mathbf{A1})$$

$$f(1) = 0, \quad f(-1) = 4 \quad (\mathbf{A1})$$

Stationary points: (1, 0) and (-1, 4)

2. Let $f(x) = 3x^2 - 6x$.

(a) Find $\int (3x^2 - 6x) \, dx$. [2]

$$\int (3x^2 - 6x) \, dx = x^3 - 3x^2 + c \quad (\mathbf{A1})(\mathbf{A1})$$

(b) Show that $f(x) = 0$ when $x = 0$ and $x = 2$. [1]

$$f(0) = 0 - 0 = 0 \quad (\mathbf{A1}) \quad (\mathbf{AG})$$

$$f(2) = 3(4) - 6(2) = 12 - 12 = 0$$

(c) Find the area of the region enclosed by the graph of f and the x -axis, between $x = 0$ and $x = 2$. [3]

$$\int_0^2 (3x^2 - 6x) \, dx = [x^3 - 3x^2]_0^2 \quad (\mathbf{M1})$$

$$= (8 - 12) - (0) = -4 \quad (\mathbf{A1})$$

$$\text{Area} = |-4| = 4 \quad (\mathbf{A1})$$

(d) A function g has derivative $g'(x) = 3x^2 - 6x$ and $g(1) = 5$. Find $g(x)$. [2]

$$g(x) = x^3 - 3x^2 + c \quad (\mathbf{M1})$$

$$g(1) = 1 - 3 + c = 5 \Rightarrow c = 7$$

$$g(x) = x^3 - 3x^2 + 7 \quad (\mathbf{A1})$$

3. A manufacturer produces items. Each item is independently classified as defective or non-defective.

- (a) In a random sample of 3 items, the probability that any one item is defective is $\frac{1}{3}$.
Let X be the number of defective items.

- (i) Write down the distribution of X . [1]

$$X \sim B\left(3, \frac{1}{3}\right) \quad (\text{A1})$$

- (ii) Find $P(X \leq 1)$. [3]

$$P(X = 0) = \left(\frac{2}{3}\right)^3 = \frac{8}{27} \quad (\text{A1})$$

$$P(X = 1) = \binom{3}{1} \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^2 = 3 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{12}{27} \quad (\text{M1})$$

$$P(X \leq 1) = \frac{8}{27} + \frac{12}{27} = \frac{20}{27} \quad (\text{A1})$$

- (b) The mass of items follows a normal distribution with mean 50 g and standard deviation σ g. It is known that 2.28% of items have mass greater than 56 g.

You are given that $P(Z > 2.00) = 0.0228$.

- (i) Find the value of σ . [2]

$$z = \frac{56 - 50}{\sigma} = 2 \quad (\text{M1})$$

$$\sigma = \frac{6}{2} = 3 \quad (\text{A1})$$

- (ii) Find $P(\text{mass} < 44)$. [2]

$$z = \frac{44 - 50}{3} = -2 \quad (\text{M1})$$

$$\text{By symmetry, } P(Z < -2) = P(Z > 2) = 0.0228 \quad (\text{A1})$$

4. Let $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find $f'(x)$. [1]

$$f'(x) = 3x^2 - 12x + 9 \quad (\mathbf{A1})$$

(b) Find the x -coordinates of the stationary points. [2]

$$3x^2 - 12x + 9 = 0 \quad (\mathbf{M1})$$

$$x^2 - 4x + 3 = 0 \Rightarrow (x - 1)(x - 3) = 0$$

$$x = 1 \text{ and } x = 3 \quad (\mathbf{A1})$$

(c) Use the second derivative to determine the nature of each stationary point. [3]

$$f''(x) = 6x - 12 \quad (\mathbf{A1})$$

$$f''(1) = 6 - 12 = -6 < 0 \Rightarrow \text{local maximum} \quad (\mathbf{R1})$$

$$f''(3) = 18 - 12 = 6 > 0 \Rightarrow \text{local minimum} \quad (\mathbf{R1})$$

(d) Find the y -coordinates and hence state the local maximum and local minimum of f . [2]

$$f(1) = 1 - 6 + 9 + 1 = 5 \quad (\mathbf{A1})$$

$$f(3) = 27 - 54 + 27 + 1 = 1 \quad (\mathbf{A1})$$

Local max at (1, 5), local min at (3, 1)